

Microprocessor System Lab6 Check List

- **Date:** 2024/11/5
 - **Please follow the instructions, and report to TAs after finishing check points.**
 - **If you have any questions or troubles during the experiment, please feel free to ask TAs or the teacher, discussion with classmates is also encouraged.**
 - **Please refer to the content from class slides, you can also use the resources from the internet.**
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Part I. Question

Please refer to the class content and answer the following questions.

Check Points:

1.1: Please provide an explanation for the instructions listed below.

MOV A, #10H

MOV A, #10

MOV A, R5

MOV A, 10H

MOV A, @R1

MOVC A, @A+DPTR

Part II. Example Projects

There are three example projects in this lab. Please refer to the source code on the google drive and execute it to see the results.

The first example involves creating the necessary delay to produce various notes. With the delay we have generated, a square wave signal with the desired period is sent to buzzer, which is already implemented on the board. As a result, the buzzer will produce specific frequency sounds, such as Do, Re, and Mi

The second example shows how to use assembly-code-based delay function to control the led.

The third example use assembly code to generate frequency. Additionally, pressing buttons triggers sound production, and LED displays different patterns corresponding to the pressed buttons.

- 2.1:** Run the first example, and understand how the programs work.
- 2.2:** Run the second example, and understand how the programs work.
- 2.3:** Run the third example, and understand how the programs work.



Please refer to the class content and example project source code to complete the Practice.

Try to implement the song ‘Twinkle, Twinkle, Little Star’ with pure assembly code. The music sheet is provided below. You have the flexibility to decide the time interval between notes to ensure that the song sounds smooth.

Note	Do	Re	Mi	Fa	So	La	Si
Frequency (Hz)	1047	1175	1319	1397	1568	1760	1976

Order	1	2	3	4	5	6	7
Note	Do	Do	So	So	La	La	So
Duration (s)	0.5	0.5	0.5	0.5	0.5	0.5	1

Order	8	9	10	11	12	13	14
Note	Fa	Fa	Mi	Mi	Re	Re	Do
Duration (s)	0.5	0.5	0.5	0.5	0.5	0.5	1

Check Points:

3.1: Play a part of 'Twinkle, Twinkle, Little Star'.